Chapter 3: Research Design Framework

Variables, Designs, Hypotheses, and Descriptive Statistics

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Overview

Before You Begin

This is the most important reading of the term. It introduces **ALL** the terminology you need for your Research Plan. Take your time — everything connects.

This reading covers:

- Independent and dependent variables
- Design types (between-subjects, within-subjects, correlational)
- Writing hypotheses (one-tailed, two-tailed, null)
- Calculating and interpreting descriptive statistics (mean, SD)

All concepts are illustrated using the DataLab tasks you completed last week.

Part 1: The Building Blocks — Variables

What Is a Variable?

A **variable** is anything that can vary — any characteristic, number, or quantity that can take different values.

In the DataLab:

- Memory scores varied (some people remembered 4 words, others 8)
- Mood ratings varied (some felt 3/10, others 8/10)
- Reaction times varied (some caught the ruler quickly, others slowly)

Independent and Dependent Variables

In experimental research, we distinguish between two types of variables:

Independent Variable (IV): What the researcher **manipulates** or controls — the presumed cause.

Dependent Variable (DV): What the researcher **measures** — the outcome that may be affected.

The DV **depends on** the IV. We're asking: Does the outcome depend on the condition?

DataLab Example: Memory Task

In the memory experiment:

- **IV**: Word type (abstract vs concrete)
- **DV**: Number of words recalled (0-10)

Lab A saw abstract words (justice, freedom, wisdom). Lab B saw concrete words (chair, river, mountain). The researcher manipulated which words you saw — that's the IV.

What we measured — how many words you recalled — is the DV. We wanted to know if recall **depended on** the word type.

When There Is No IV

Not all research involves manipulation. Sometimes we simply **measure** two things and look at their relationship.

In the Frontal Lob task, you rated how good you thought you'd be at throwing (1-10), and then we measured your actual accuracy. Neither was manipulated — we just measured both.

! Key Insight

When there's no manipulation, there's **no IV**. This is called a **correlational design**, and we'll cover it shortly.

Data Types

The DV can be measured in different ways:

Data Type	Description	Example
Continuous	Can take any value (including decimals)	Reaction time (245.3ms)
Discrete	Whole numbers only (counts)	Words recalled (7)
Categorical	Categories with no numerical order	Lab group (A or B)
Ordinal	Categories with an order	Rating (1-10 scale)

In the DataLab:

- Memory score is **discrete** (you can't recall 6.5 words)
- Reaction time is **continuous** (could be 245.3ms or 245.4ms)
- Mood rating is **ordinal** (1-10 scale with order, but intervals may not be equal)

Part 2: Design Types

The **design** refers to how participants are assigned to conditions.

Between-Subjects Design

Between-subjects design: Different participants in each condition.

In the memory experiment, you were in **either** Lab A (abstract words) **or** Lab B (concrete words) — never both. Different people were in each condition.

Why "between"? We're comparing **between** groups — Lab A people versus Lab B people.

Advantage: No order effects (practice, fatigue, boredom)

Disadvantage: Individual differences may affect results (maybe Lab B just happened to have better memorisers)

Within-Subjects Design

Within-subjects design: The same participants measured in all conditions.

In the pre-post mood survey, **you** rated your mood before **and** after the session. The same person provided both data points.

Why "within"? We're comparing **within** the same person — your pre-mood versus your post-mood.

Advantage: Each person is their own control! Individual differences are controlled because we're comparing you to yourself.

Disadvantage: Order effects possible (practice, fatigue)

Within-Subjects Examples from DataLab

Task	Same Person Compared
Pre/Post mood	Before vs after session
Pre/Post energy	Before vs after session
Pre/Post stress	Before vs after session
Frontal Lob	Dominant vs non-dominant hand

Matched Pairs Design

Matched pairs design: Participants are paired on a key characteristic, then one from each pair goes to each condition.

This is a hybrid — different people in each condition (like between-subjects), but they're matched to be similar (reducing individual difference problems).

Example: Match students by the amount of sleep they reported the night before. Let's say two participants reported five hours and two participants reported 10 hours. One from each pair gets abstract words, the other gets concrete words. This would allow me to control for sleep quantity (five or ten hours), as I would have each level represented in each group. It means I would have to ask about sleep before allocating to condition, and so is a little bit more complicated in everyday life.

Correlational Design

Correlational design: Measuring the relationship between two variables — no manipulation.

When we measured self-rating and actual throwing performance, we didn't manipulate either. We simply measured both and asked: Are they related? This could be the same for hours of sleep and memory performance, or any of the personality traits we measured and any other outcome variable.

> ⚠ Critical Point
>
> In correlational designs, there is **NO IV**. Neither variable is the "cause" — we're just measuring a relationship.

Summary: Design Types

Design	Key Feature	Example
Between-subjects	Different people in each condition	Lab A vs Lab B
Within-subjects	Same people in all conditions	Pre vs Post mood
Matched pairs	Participants paired, one per condition	Matched by ability
Correlational	Relationship measured, no manipulation	Self-rating vs performance

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Part 3: Writing Hypotheses

A **hypothesis** is a prediction about what you expect to find.

Experimental vs Null Hypothesis

Experimental hypothesis (H_1 or Ha): Predicts an effect or difference will exist.

Null hypothesis (H_0): Predicts no effect or no difference.

For the memory experiment:

- **H_1**: Concrete words will be easier to remember than abstract words.
- **H_0**: There will be no difference in recall between word types.

Statistical testing works by assuming H_0 is true, then checking if the data is surprising enough to reject that assumption.

One-Tailed vs Two-Tailed

One-tailed hypothesis: Predicts the **direction** of the effect.

Two-tailed hypothesis: Predicts a difference but **not** which direction.

One-tailed examples:

- "Concrete words will be **easier** to remember than abstract words" (predicts concrete > abstract)
- "Mood will **improve** after the session" (predicts post > pre)
- "Dominant hand will be **more accurate** than non-dominant hand" (predicts dominant > non-dominant)

Two-tailed examples:

- "Word type will **affect** memory" (could be better or worse)
- "Mood will **change** after the session" (could go up or down)
- "There will be a **difference** in accuracy between hands" (no direction predicted)

When to Use Which?

Use **one-tailed** when:

- Previous research strongly suggests a direction
- Your theory predicts a specific direction
- You only care about effects in one direction

Use **two-tailed** when:

- You're uncertain about the direction
- Effects in either direction would be interesting
- You're being conservative (harder to reach significance)

Hypotheses for Each DataLab Task

Task	One-tailed Hypothesis
Memory	Concrete words will be recalled better than abstract words
Pre-Post Mood	Mood will improve after the session
Frontal Lob	Dominant hand will be more accurate than non-dominant
Sixth Sense	Detection accuracy will be greater than chance (50%)

Part 4: Descriptive Statistics

Before running statistical tests, we need to **describe** our data. Descriptive statistics summarise the key features.

The Mean

The **mean** is the arithmetic average — the most common measure of central tendency.

$$\bar{x} = \frac{\sum x_i}{n}$$

Calculating by hand:

1. Add up all values (Σx)
2. Divide by the number of values (n)

Example: Memory scores: 5, 7, 6, 8, 9

- Sum: 5 + 7 + 6 + 8 + 9 = 35
- Count: 5
- Mean: 35 ÷ 5 = **7**

The Standard Deviation

The mean alone doesn't tell the whole story. Two groups can have the same mean but look very different:

- Group A: 6, 6, 7, 7, 8 (Mean = 6.8)
- Group B: 2, 5, 7, 9, 11 (Mean = 6.8)

Group B is much more spread out. We need to measure this **variability**.

Standard Deviation (SD) measures how spread out the data is from the mean.

- Low SD = scores clustered together (consistent)
- High SD = scores spread out (variable)

Calculating SD by Hand

Formula:

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

Steps:

1. Calculate the mean
2. Subtract the mean from each value (deviation)
3. Square each deviation
4. Sum the squared deviations
5. Divide by (n-1)
6. Take the square root

Example: Scores: 5, 7, 6, 8, 9 (Mean = 7)

Score	Deviation (x - mean)	Squared
5	5 - 7 = −2	4
7	7 - 7 = 0	0
6	6 - 7 = −1	1
8	8 - 7 = +1	1
9	9 - 7 = +2	4
Sum		**10**

- Variance = 10 ÷ (5-1) = 2.5
- SD = √2.5 = **1.58**

Why n-1?

We divide by (n-1), not n, because we're estimating the population SD from a sample. This "degrees of freedom" correction makes our estimate unbiased.

Interpreting SD

For normally distributed data:

- ~68% of values fall within 1 SD of the mean
- ~95% fall within 2 SDs
- ~99.7% fall within 3 SDs

If the mean memory score is 7 with SD = 1.58:

- About 68% of people scored between 5.4 and 8.6
- About 95% scored between 3.8 and 10.2

The Key Insight

> ! Signal vs Noise
>
> A difference between means only matters if it's **big relative to the spread**.
>
> - **Signal** = difference between group means
> - **Noise** = variability within groups (SD)
> - The key question: Is the signal **bigger** than the noise?

Part 5: Reporting Descriptive Statistics

When writing up research, report descriptives clearly using APA format.

Basic Format

Report both mean AND standard deviation:

> (*M* = value, *SD* = value)

Example:

> Participants' memory scores were moderate (*M* = 6.8, *SD* = 1.58).

Comparing Groups

When comparing conditions, report descriptives for each:

> Lab A participants (*M* = 5.8, *SD* = 1.4) recalled fewer words than Lab B participants (*M* = 6.5, *SD* = 1.3).

APA Conventions

- Use italics for statistical symbols: *M*, *SD*, *t*, *p*, *r*, *d*
- Report two decimal places
- No leading zero for values that cannot exceed 1 (e.g., *p* = .023, not *p* = 0.023)

Putting It All Together: The DataLab Framework

Each DataLab task illustrates different design elements:

Task	Design	IV	DV
Memory	Between	Word type	Recall (0-10)
Pre-Post Mood	Within	Time	Rating (1-10)
Pre-Post Energy	Within	Time	Rating (1-10)
Pre-Post Stress	Within	Time	Rating (1-10)
Frontal Lob (hands)	Within	Hand	Accuracy (%)
Sixth Sense	vs Chance	None	Hits (0-6)
Self-rating vs Performance	Correlational	None	—
Extraversion vs Outcomes	Correlational	None	—

The Research Design Decision Tree

When planning or analysing research, follow this logic:

```
Research Question
       ↓
Can you manipulate the IV?
       ↓
   ┌───┴───┐
  Yes      No
   ↓        ↓
EXPER    CORRELATIONAL
   ↓
Same or
different
people?
   ↓
┌──┴──┐
Same  Different
  ↓       ↓
WITHIN  BETWEEN
```

Check Your Understanding

Before the lecture, make sure you can answer:

1. What's the difference between an IV and a DV?
2. When would you use a between-subjects design vs within-subjects?
3. What's the difference between a one-tailed and two-tailed hypothesis?
4. Why is there no IV in a correlational design?
5. If you're comparing pre-test and post-test scores from the same participants, which design type is it?
6. Calculate the mean of: 4, 6, 8, 10, 12
7. Why do we report SD alongside the mean?

Summary

Concept	Key Point
IV	What the researcher manipulates
DV	What the researcher measures
Between-subjects	Different people in each condition
Within-subjects	Same people in all conditions
Correlational	No manipulation, no IV
One-tailed	Predicts direction
Two-tailed	Predicts difference, not direction

Concept	Key Point
H_0	Predicts no effect
Mean	Average value
SD	Spread from the mean

Connecting to Assessments

This reading prepares you for **ALL** assessments:

Assessment	Relevant Content
Research Plan	IV/DV identification, design type, hypothesis writing
MCQ Exam	Design identification, terminology
EDS Assignment	Calculating mean and SD by hand, identifying designs
Research Report	Method section structure, reporting descriptives

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Further Reading

For additional depth:

- Coolican, H. (2017). Chapters 3 and 15: Research design and descriptive statistics. VLeBooks